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EXPERIMENTAL INVESTIGATION OF THE CHEMISTRY OF EXCITED STATES OF RARE GASES

Technical Progress Report
for Oct. 15, 1975 - July 15, 1976

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Abstract

Total quenching rate constants and branching ratios for rare gas halide excimer formation have been measured for $Ar(^3P_{0,2})$, $Kr(^3P_2)$ and $Xe(^3P_2)$ reacting with chlorine and fluorine-containing molecules. Emphasis was focused on systems giving KrF* and XeF* as products. All halogen containing molecules quench metastable rare gas atoms with gas kinetic cross-sections. However, only diatomic halogens and small molecules with O-X or N-X bonds quench the metastable with high branching ratios for excimer formation. The product channels for the reagents which do not yield the rare gas halides as the main exit channel remain to be identified. The work that has been done provides a good survey of halogen donors. This work will assist in selection of the best halogen-containing molecules for use in rare gas halide excimer lasers.

Tunable dye laser fluorescence experiments have been done to obtain radiative lifetimes and two body quenching rate constants (with ground state Ar) for excited states of $Ar^*(3p^5,4p)$ and $Ar^*(3p^5,5p)$.

Scope of Investigations:

The main objectives of the proposed experimental work for the 1975-1976 contract year were the following:

(a) Characterization of the interactions between metastable states of Ar, Kr and Xe atoms with halogen containing molecules.

(b) Develop the pulsed, tunable dye laser technique into a tool that could be used to study processes occuring in the metastable atom flowing-afterglow systems.

(c) Initiate the development of new experimental methods for studying the chemistry of excited atomic and molecular states of the rare gases.

Satisfactory progress has been made on points (a) and (b), and manuscripts will have been submitted by Oct. 15, 1976 to appropriate journals describing portions

of the work. Considering the fact that funds actually were not made available until Dec. 15, 1975, the degree of progress is unusually good; this can be attributed to the efforts of Velazco and Rodriguez. Little has been done under category (c), because of the success with (a) and (b) and because of lack of additional personnel. However, preliminary work using a microwave powered resonance lamp to study the sensitized reactions of $Kr(^1P_1)$ and $Kr(^3P_1)$ looks promising. We also are attempting to put a small vacuum U.V. spectrometer into operation. It will be used with the flowing afterglow to monitor excitation processes between 150-200 nm. Funds are requested in the proposal for 1976-1977 for purchase of a better vacuum U.V. monochromator.

Summary of Personnel:

The personnel supported by this contract consisted of one graduate student, Mr. Enrique Velazco, one postdoctoral fellow, Dr. Jose Rodriquez, and the principal investigator, Dr. D. W. Setser. Mr. Velazco and Dr. Rodriquez spent 100% of their time on this contract. The principal investigator received one month's summer salary from this contract, and he spent approximately 25% of his research time working on the ERDA contract. Mr. Velazco will be writing his Ph.D. Dissertation during the July 15 - Oct. 15 quarter.

Research Results:

(i) Rare gas halide excimer formation from reactions of $Kr(^3P_2)$ and $Xe(^3P_2)$ atoms with halogen-containing molecules.

Total quenching rate constants and the branching ratios for formation of KrF*, KrCl*, XeF* and XeCl* have been measured for reactions of $Kr(^3P_2)$ and $Xe(^3P_2)$ with fluorine and chlorine containing molecules. This work was thoroughly described in the second quarterly progress report, which contained the preprint of the manuscript submitted to J. Chem. Phys. A concise summary

of the work also is presented in Appendix I of this report, which is the abstract of the paper presented at the Twelfth Informal Photochemistry Conference, Washington, D.C.

Reactive quenching of $Xe(^3P_2)$ and $Kr(^3P_2)$ with formation of the chloride or fluoride excimer occurs with unit efficiency for only Cl_2, F_2, ClF, OF_2, NF_3 and perhaps NOCl. For larger molecules containing F or Cl attached to other elements by weak bonds, e.g., even CF_3OF, the excimers are formed but with reduced branching ratios. Interhalogen compounds usually give excimers of both halogens with the KrF* or XeF* channel being less favored. The best donor molecules for laser applications certainly appear to be the diatomic halogens; although, for specialized applications where corrosion or toxicity are problems, other donors may be used. The work done in our laboratory should be sufficient for selecting the best donor for a given laser application.

In general terms, the charge-transfer mechanism appears to describe the mechanism for reactive quenching. There are, however, <u>many</u> details yet to be explained, e.g., the various excimer product ratios found for the interhalogen compounds, the product channels for cases in which excimer formation is not important, etc. Mr. Velazco will address some of these questions during the course of writing his dissertation. We also are interested in trying to better understand the bound-free emission spectra from the highly vibrationally excited excimers.

During the course of our search for good fluorine donor molecules, experiments were done with XeF_2. Quenching of metastable Ar*, Kr* and Xe* by XeF_2 gives emission from both $XeF(^2\Sigma^+)$ and $XeF(^2\Pi_{1/2})$. We are in the process of trying to obtain high quality spectra of the $(^3\Pi_{1/2}-^2\Sigma^+)$ transition; these data will be shared with Dr. Tellinghuisen to better facilitate the characterization of the XeF(X,B and D) states.

During the next few months, experiments will be done to obtain the rate constants for ArF* formation from reaction of $Ar(^3P_{0,2})$ with RF donors.

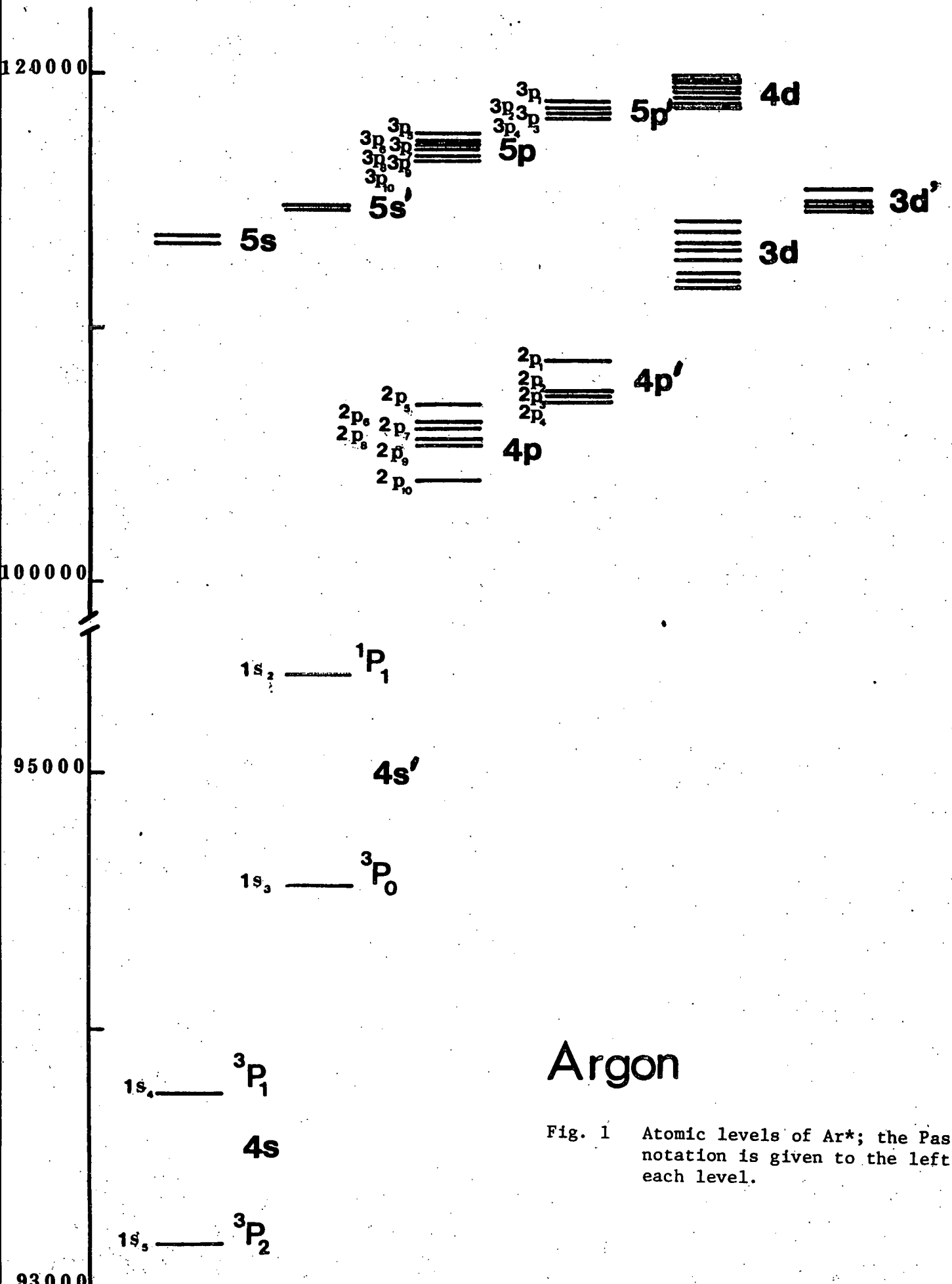
These results will be compared to the formation rate constants for the KrF* and XeF* systems. In order to do the ArF* studies, it is necessary to use a vacuum monochromator since the emission covers the 175-230 nm region. The ArF* studies are especially important to confirm some assumptions made in assignment of unity branching ratios for formation of XeF* and KrF* from F_2.

(ii) Lifetimes and two-body quenching rate constants of Ar*(4p) and Ar*(5p) states.

This work has not been described in previous quarterly reports. Therefore a fairly detailed discussion follows.

Introduction

Collisional quenching rate constants and radiative lifetimes of excited states of argon and other rare gases are important for understanding the energy transfer steps pertinent to the operation of electrically pumped lasers using argon as the principal gas. The direct excitation of individual excited states of argon with a pulsed light source followed by observation of the fluorescence is a convenient method for the determination of lifetimes and quenching rates. A tunable dye laser pumped by a nitrogen laser provides this high intensity source. This source in combination with a hollow cathode discharge flow system enables us to optically pump the $Ar(4s,{}^3P_{0,2})$ metastable states to the higher 5p and 4p states. The atomic levels of interest are shown in Figure 1. The notation of the excited rare gas atomic states is somewhat confusing. The four states arising from the $Ar(3p^5 4s)$ configuration are usually named according to L-S coupling as ${}^3P_2$, ${}^3P_1$, ${}^3P_0$, ${}^1P_1$ in order of increasing energy. The more proper J_c-1 coupling notation is $4s[3/2]_2$, $4s[3/2]_1$, $4s'[1/2]_0$ and $4s'[1/2]_1$; whereas, the Paschen numbering is $1s_5$, $1s_4$, $1s_3$, and $1s_2$. The higher states are normally identified by the J_c-1 notation or the Paschen numbering. The present report describes the experimental technique, as well as some results obtained by the excitation of the Ar*(5p) and Ar*(4p) states. No interpretation of the data will be attempted.



Argon

Fig. 1 Atomic levels of Ar*; the Paschen notation is given to the left of each level.

Experimental

The apparatus consisted of our usual hollow-cathode discharge flow system for generating $Ar(^3P_2)$ and $Ar(^3P_0)$ atoms at a concentration of $\sim 5 \times 10^{10}$ atoms cm^{-3}. The metastable atoms, highly diluted in Ar, flowed into a laser induced fluroescence cell that had two baffle arms for the entrance and exit of the laser light and two viewing windows. One window was backed with a mirror and the other with a suitable filter to isolate the laser induced fluorescence. A RCA 7265 photomultiplier tube viewed the fluorescence zone through various filters chosen to best isolate the laser induced fluorescence. The signal of the photomultiplier tube was averaged by a PAR 162 boxcar integrator with two model 164 channel modules. One channel (B) observed only the peak of the fluorescence signal; while the other (A) scanned and averaged the signal as a function of time. In order to compensate for fluctuations in the incident laser intensity, the output of channel A over channel B was recorded. The laser system was a homemade Hansch type dye laser pumped by an AVCO 250 kwatt nitrogen laser. The laser was operated at 30 to 50 pps; the bandwidth was less than 0.1 Å and the pulse duration was ~ 5 nsec. The trigger circuit from the power supply module was used to trigger the boxcar and the oscilloscope. The desired wavelength from the dye laser was selected by adjustment of the grating forming one end of the dye laser cavity and monitored with a 1/2 meter Jarrel Ash monochromator.

For the $Ar^*(5p)$ states, the scintillators 0.001 M 1,4 bis[2-(5-phenyl-oxazolyl)benzene in p-dioxane and 7×10^{-4} M 1,4 bis[2-(4-methyl-5-phenyl-oxazolyl) benzene in p-dioxane were found to be appropriate dyes for the spectral range, 407.0 - 451.5 nm and 418.0 - 465.0 nm, respectively. These dyes are commonly called POP and dimethyl-POPOP. A mixture of rhodamine-5G and cresyl violet perchlorate in ethanol was used as the dye for the study of the $Ar^*(4p)$ states. Only the three highest energy states of the 4p manifold could be studied because of the long wavelength limitation of the dye; the wavelength range was from 714.7 to 696.5 nm.

Results and Analysis

A. Argon 4p states.

The experiments with the Ar*(4p) levels were very straightforward because only the fluorescence from the level that was pumped was monitored. The transitions from the $2p_2$, $2p_3$, and $2p_4$ levels to $4s'[1/2]_1$ following laser excitation was isolated from other transitions with an interference filter. The fluorescence decay curves were simple exponentials and were analyzed to yield pseudo first order rate constants. These decay curves were obtained for a range of pressure, and a plot of the observed decay constants vs argon pressure are shown in Figures 2 for the $4p'[3/2]_1$ level. Such plots were fitted to a least squares line, and the zero pressure intercepts and slopes are tabulated in Table 1. The standard error limits are ±15%. Comparison of the lifetimes of the three Ar*(4p) states, which is the inverse of the zero pressure intercepts, to other measured and calculated lifetimes are given below:

Lifetimes of some 4p Levels

Level	λnm	Present Work (n sec)	Previous Work (n sec)			
			(a)	(b)	(c)	(d)
$2p_2$	696.5	33.5 ± 4	25 ± 2	25.7	24 ± 3	---
$^2p_3$	706.7	36.3 ± 4	26 ± 1	27.6	25 ± 3	---
$^2p_4$	714.7	45.2 ± 5	31 ± 2	29.0	27 ± 3	37 ± 1

(a) J. Z. Kloze, J. Opt. Soc. Am. 58, 1200 (1968).

(b) R. A. Lilly, J. Opt. Soc. Am. 66, 245 (1976). (calculated values)

(c) J. B. Shumaker, Jr., and C. H. Popenone, J. Opt. Soc. Am. 57, 8 (1967).

(d) J. P. Grandin, D. Lecler, J. Margerie, Acad. Sci. (Paris) C. R. (A&B) 272, 929 (1971). In addition to this absolute measurement there is fragmentatory evidence, usually based upon oscillator strengths of a particular line, that also favor longer lifetimes than given by a,b, and c.

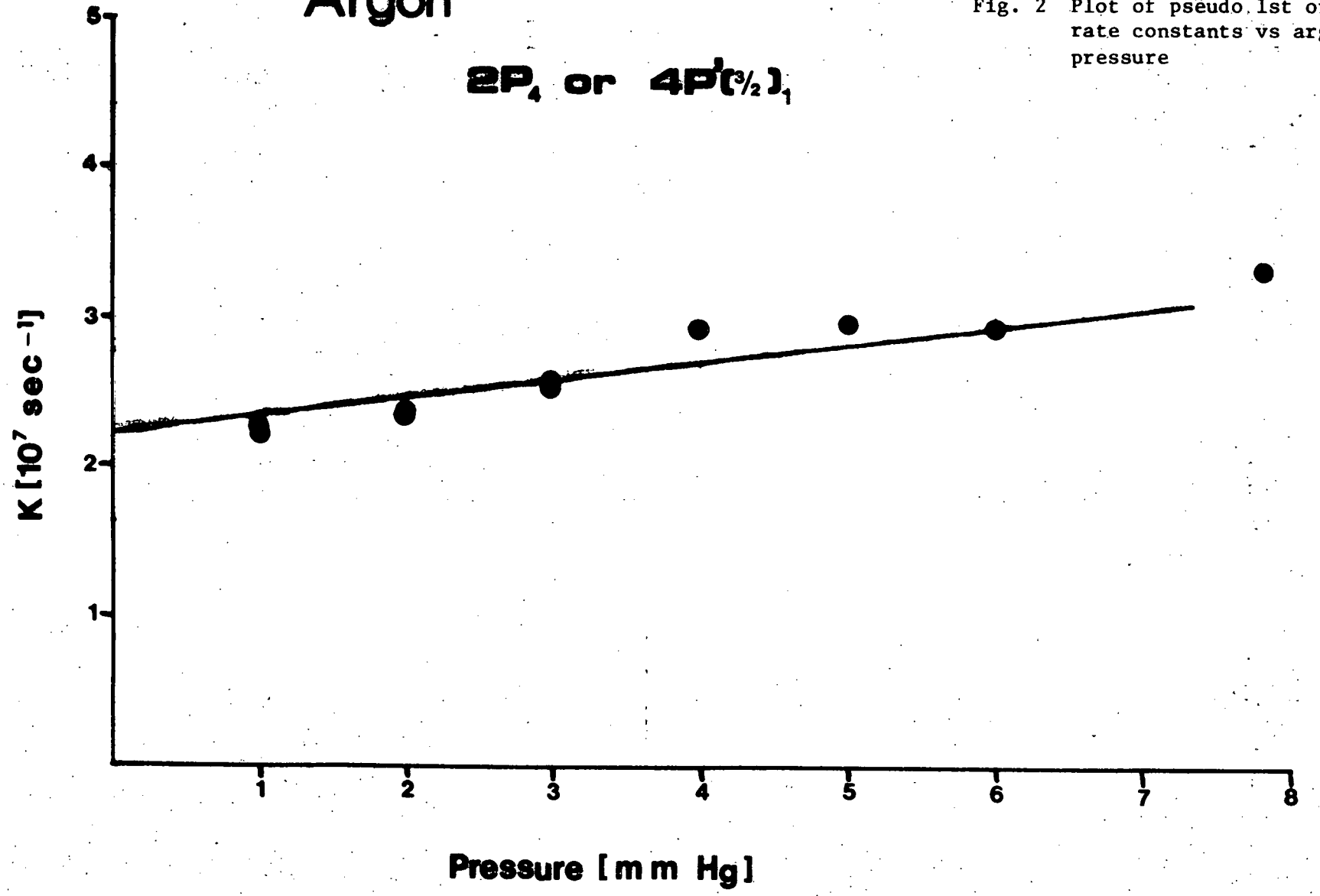


Fig. 2 Plot of pseudo 1st order rate constants vs argon pressure

Table 1 Argon 4p States

Level	laser wavelength[a] nm	k_o or τ^{-1} (sec^{-1})	Rate Constant (cm^3 $atom^{-1}$ sec^{-1})
$2p_2$ or $4p'[1/2]_1$	696.54	2.92×10^7	5.21×10^{-11}
$2p_3$ or $4p'[3/2]_2$	706.72	2.74×10^7	6.62×10^{-11}
$2p_4$ or $4p'[3/2]_1$	714.70	2.22×10^7	3.69×10^{-11}

a. In each case the upper level was produced by pumping from the lower metastable Ar(4s) level, see Fig 1.

Table 2 Argon 5p States

Level	laser wavelength[a] nm	k_o or τ^{-1} (sec^{-1})	Rate Constant (cm^3 $atom^{-1}$ sec^{-1})
$3p_6$ or $5p[3/2]_2$	415.86	0.78×10^7	5.15×10^{-11}
$3p_7$ or $5p[3/2]_1$	416.42	0.83×10^7	5.89×10^{-11}
$3p_8$ or $5p[5/2]_2$	419.07	0.81×10^7	5.15×10^{-11}
$3p_9$ or $5p[5/2]_3$	420.07	0.85×10^7	5.62×10^{-11}
$3p_{10}$ or $5p[1/2]_1$	425.12	0.75×10^7	5.75×10^{-11}
$3p_2$ or $5p'[1/2]_1$	418.2	0.16×10^7	3.74×10^{-11}

The $3p_6$ - p_{10} levels were produced by pumping from the lower metastable Ar($^3P_2$) level, see Fig 1. The p_2 level was produced by pumping from the upper metastable, Ar($^3P_0$), level.

The difference between our values and the others appear to be outside the random error limits and suggest a systematic difference. As noted by Wiese Smith and Miles in their NBS tabulation, the experimental lifetime data tend to fall into two groups. Our data appear to support the longer lifetime measurements. These Ar*(4p) studies were our first experiments and several improvements in apparatus and technique subsequently have been made. Therefore, we intend to reinvestigated these levels over a wider pressure range. There is no apparent reason why the uncertainty cannot be reduced to $\pm$5%. If there is systematic error, it should affect the intercepts more than the rate constants. Since a single Ar*(4p) level was monitored, the two body quenching rate constants are for all collision processes that remove atoms from the particular level being observed. The main processes presumably are collisional transfer to the lower energy levels of the Ar*(4p) manifold. Experiments to identify product states by observation of emission from other Ar*(4p) levels were unsuccessful. One of the difficulties was the background emission generated by the hollow-cathode discharge, which contributes to the noise level observed by the boxcar. With our recent improvements, it may be possible to identify the individual Ar(4p) product states. This possibility will be checked in our reinvestigation of these levels.

B. Argon 5p states

Pumping the Ar*(5p) levels with the laser is easier than the Ar*(4p) levels because of the superior performance of the dye laser in this wavelength range. However, despite considerable effort we were unable to observe the laser induced fluorescence from the excited Ar*(5p) level back to the Ar($^3P_1$ or $^1P_1$) levels. In fact, the 5p-4s fluorescence was not even observable with band-pass filters that transmitted all of the emission from the 5p-4s transitions. Therefore, an indirect observation scheme was adopted. This is demonstrated in Figure 3. The observed fluorescence

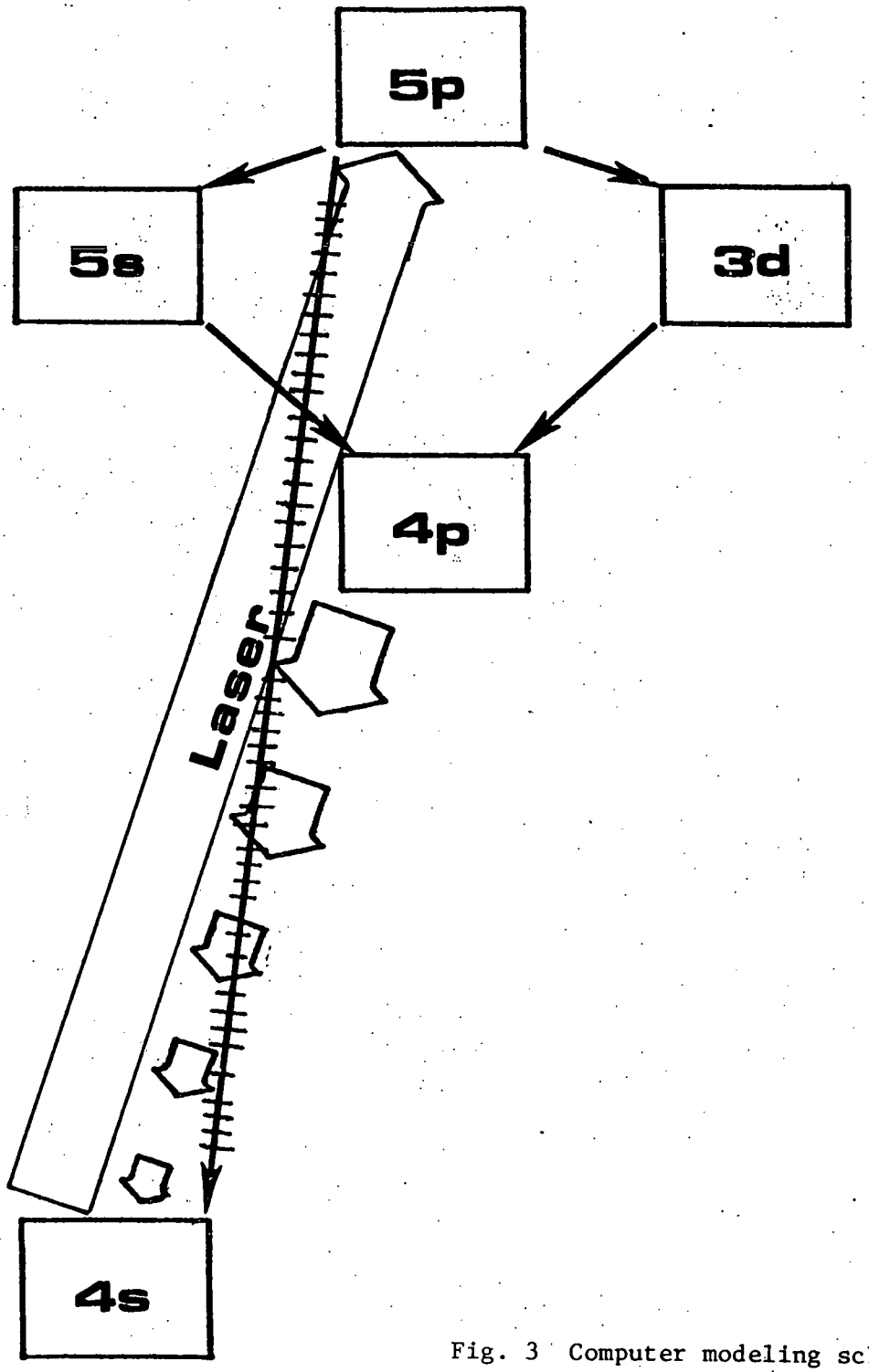


Fig. 3 Computer modeling scheme used to analyze the 5p states. The 5p-4s transitions could not be observed.

was transitions from the 4p levels to the 4s levels at long wavelengths. This long wavelength emission intensity was monitored with a wide band-pass filter, which permitted the photomultiplier to view all emission originating from the 4p levels.

Figure 4 shows a typical decay plot obtained by monitoring the emission intensity following excitation of $Ar^*(5p[3/2]_2)$. Detailed computer modeling of the process indicated in Figure 3 was done by considering the equations below.

$$Ar^*(5p) \xrightarrow{\tau^{-1}} Ar^*(3d \text{ and } 4s) + h\nu$$

$$\xrightarrow{k_Q} Ar^*(3d \text{ and } 5s)$$

$$Ar^*(3d \text{ and } 5s) \xrightarrow{\tau^{-1}} Ar^*(4p) + h\nu$$

$$Ar^*(4p) \xrightarrow{\tau^{-1}} Ar^*(4s) + h\nu$$

Collisional quenching of the $Ar^*(3d + 5s)$ and $Ar^*(4p)$ can be included in the scheme; however, this has no affect upon interpretation of the long-term decay component of Figure 4. Computer modeling of the reaction scheme indicated that the decay curve was controlled by the kinetics of the 5p states, because the radiative lifetimes of the $Ar^*(3d \text{ and } 5s)$ and $Ar^*(4p)$ are shorter than for $Ar^*(5p)$. The long-term decay was simply exponentail and the first order rate constants, derived from just the long-term portion of the decay curves, are shown in Figure 5 for $Ar^*(5p[5/2]_2)$. Extracting the rate constant from the long-term part of the decay curves at higher pressure is not straightforward because of the increase in magnitude of the rate constant and because of poor signal to noise ratio at the higher pressures. The failure of the simple analysis is indicated by the low values of the first order rate constants at higher pressures shown in Figure 6. Table 2 (see page 10) summarizes the collisional quenching rate constants ($5p \rightarrow (5s + 3d)$ levels) and the radiative lifetimes of the 5p states. These results are based upon fitting the data at pressure below 8 torr to least squares straight lines.

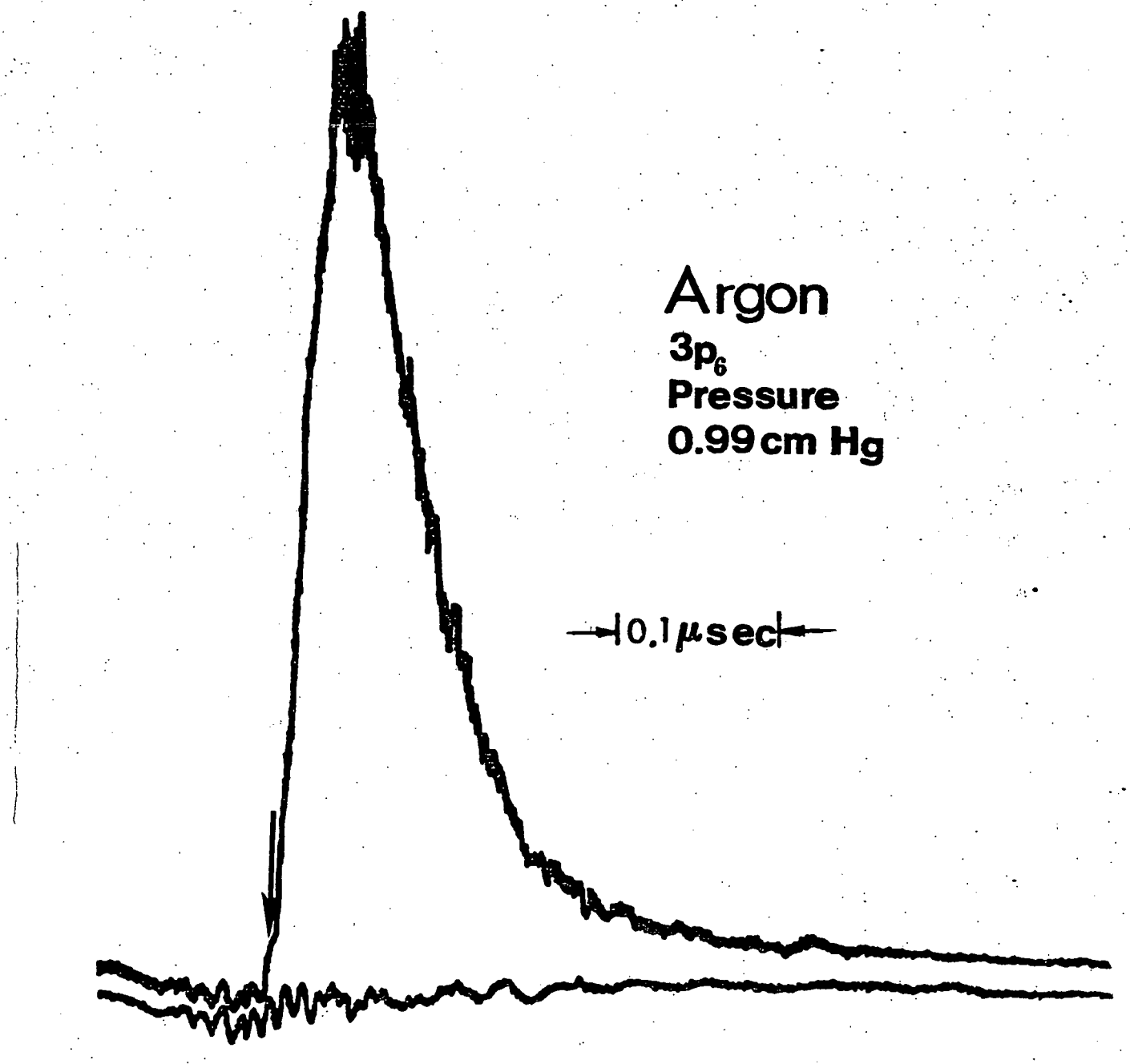


Fig. 4 The arrow denotes the location of the laser pulse, which is zero time. The delay between the laser pulse and the maximum in the fluorescence signal was ∿50n sec.

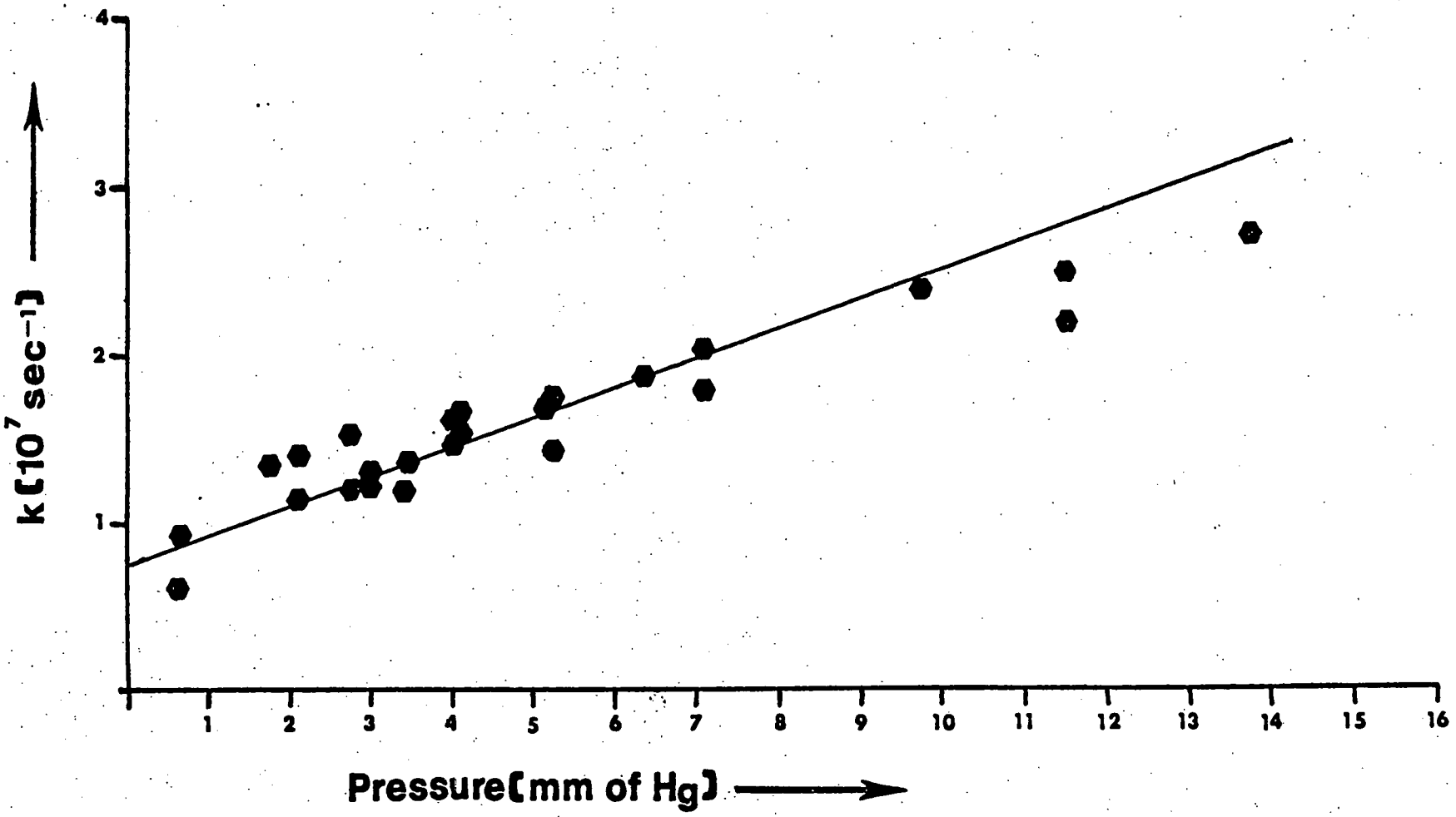


Fig. 5 Plot of Pseudo First Order Rate Constants vs Argon Pressure

The common value of the two-body rate constants implies that collisional relaxation within the $Ar^*(5p)$ levels occurs prior to transfer to the 5s and 3d levels. Thus, the measured values are the rate constant for transfer from the manifold of 5p levels to the 3d and 5s manifolds. By fitting the entire decay plot, rather than using just the long-term portion, it should be possible to extract a more reliable two-body quenching rate constant. We fully expect this more complete data treatment to restore the points in the 8-16 torr range to the linear plot. Based upon the data in hand, we see no evidence for three-body quenching that would effectively compete with the two-body transfer from the 5p manifold to the 5s and 3d manifolds. An attempt will be made to model the rise time of the decay plots, see Fig. 4, to extract information about the processes converting the 3d + 5s levels to the 4p levels. The <u>failure</u> to observe fluorescence from the 5p levels back to the 4s levels is particularily <u>hard to understand</u>. According to the calculated radiative branching ratios for the transitions from the 5p levels to the 5s, 3d and 4s levels, the emission from $Ar^*(5p)$—$Ar^*(4s)$ should have been observable. The intercept values from the $Ar^*(5p)$ experiments are very close to the calculated radiative lifetimes. The tentative explanation, very fast collisional transfer within the $Ar^*(5p)$ manifold, is not totally acceptable. As an experimentalist, another tempting explanation is that the calculated branching ratios may be somewhat in error. Further work remains to be done in order to understand these apparently "simple" experiments.

Technical Reports, Preprints, Conference Abstracts

1) First Quarterly Report C00-2807-1

2) Second Quarterly Report C00-2807-2

3) Abstract of Paper presented at the Twelfth Informal International Photochemical Conference (attached).

4) Manuscript submitted to J. Chem. Phys. (included as part of the C00-2807-2 report).

Appendix I

Abstract of Paper presented at Twelfth Informal Photochemistry Conference.

QUENCHING RATE CONSTANTS FOR $Ar(^3P_2)$, $Kr(^3P_2)$ and $Xe(^3P_2)$ BY HALOGEN-CONTAINING MOLECULES, AND BRANCHING RATIOS FOR XeF AND KrF FORMATION*

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The discovery[1,2] of rare gas-halogen excimer formation from the reaction of rare gas metastable atoms with halogen or halogen-containing molecules, has led to the development of a new class of ultraviolet lasers.[3] However, scaling-up of these prototype devices faces some practical as well as theoretical problems: halogen donors must be found that give excimer formation with high efficiency, do not absorb at the laser frequency and are reasonably easy to handle. Modeling of these systems requires the knowledge of the rate constants (and branching ratios) for the relevant steps. In this work total (thermal) quenching rate constants for $Ar(^3P_2)$, $Kr(^3P_2)$ and $Xe(^3P_2)$ with a variety of fluorine donor reagents are reported; also, the branching ratios for the formation of KrF* and XeF* were measured. Emission from a second excimer state of XeF, XeCl, KrCl and KrF, which correlated with the $(^2P_{1/2})$ state of the rare gas ion will be reported, as will the observation of vibrational relaxation of the excimers in the 1-10 torr pressure range.

Total quenching rate constants (k_Q) were measured using a flowing after glow apparatus[4] with a linear flow velocity of $\sim$80 $msec^{-1}$. The Ar metastable atoms were produced by passing a flow of purified Ar through a hollow cathode discharge. Total metastable concentrations $\sim 5 \times 10^{10}$ cc^{-1} were measured at the first observation point, with $Ar(^3P_2)$ and $Ar(^3P_0)$ in a $\sim$6:1 ratio. The $Kr(^3P_2)$ or $Xe(^3P_2)$ metastables were produced by energy transfer reactions from $Ar(^3P_{0,2})$ by adding small quantities ($\sim$0.3%) of Kr or Xe to the Ar flow

before the discharge. The decay of the metastables along the tubular reactor as a function of quencher concentration was followed by atomic absorption using Pen Ray source lamps: the Ar 811.5 nm, Kr 811.2 nm, Xe 881.9 nm lines are used. Fluorine donors were diluted in Ar and stored in a passivated stainless vessel connected to the inlet part of the flow reactor by 1/4" ss tubing. Less reactive reagents were stored and metered through glass lines. The quenching rate constants, k_Q, from these experiments are listed in Table I. The reproducibility of k_Q for the highly corrosive F-donors is $\pm$20% due mainly to difficulties in purification of samples. For chlorine and fluorides that do not attack glass, the reproducibility is $\pm$ 10%.

A comparison technique was developed to measure the branching ratio for excimer formation, k_{MX}/k_Q, which eliminates the need of a direct and absolute measurement of k_{MX}. In the absence of quenching or non-radiative processes, the excimer emission intensity is given by (1).

$$I_{MX}^{RX} = k_{MX}[M^*][RX] \qquad (1)$$

Hence, if k_{MX} is assigned for a reference reaction, and if experiments are done for the same [M*], we can obtain other k'_{MX} from (2),

$$k'_{MX} = k_{MX} \frac{I_{MX}^{R'X}/[R'X]}{I_{MX}^{RX}/[RX]} \qquad (2)$$

by simply comparing the relative emission intensities for known RX and RX' concentrations. In order to perform these intensity comparisons, a system was built that consists of two gas handling sections (one grease-free but of glass and the other stainless steel) and the flow reactor section. Presssures and flow rates were monitored with a pressure transducer. Emission from the reaction of $Kr(^3P_2)$ or $Xe(^3P_2)$ (produced a described above) with added halogen donor, was detected by a computer controlled 0.75 m Jarrell-Ash

monochromator equipped with a cooled photomultiplier tube and SSR photon counter. For low pressure spectra, the intensity (counts/sec) was recorded at 1 Å intervals; whereas, for high pressure spectra shorter intervals were used. The data were stored in magnetic tapes and subsequently corrected for spectral response. Integrated emission intensities were obtained by simple addition of the counts in each interval.

Molecular chlorine was selected as the reference donor (RX) because the absolute values of k_{KrCl} and k_{XeCl} could be estimated, and also because Cl_2 can be handled in a relatively simple and reliable way. Columns 4 and 5 of Table 1 list the rate constants for excimer formation relative to Cl_2 (see eq.(2)). The absolute value of k_{KrCl} from the Kr* + Cl_2 reaction was calculated by using the absolute rate constant for the formation of the Cl_2* 257 nm continuum from Ar* + Cl_2 (1.7×10^{-11} cc molec^{-1} sec^{-1}). Comparing the emission intensities of the Cl_2* 257 nm continuum and the KrCl excimer emission gives (3).

$$\frac{I_{Cl_2}^*}{I_{KCl}^*} = \frac{k_{Cl_2}^{Ar}}{k_{KrCl}} \frac{[Cl_2]}{[Cl_2]} \frac{[Ar(^3P_{0,2})]}{[Kr(^3P_2)]} \qquad (3)$$

The ratio of Ar and Kr metastable concentrations was determined and the result for equal $[Cl_2]$ was $k_{KrCl} = 52 \times 10^{-11}$ ml molec^{-1} sec^{-1}. This value gives a branching ratio of 0.7 for excimer formation from Kr* + Cl_2. This value however, may have an uncertainty of as much as $\pm$ 50% because the intensity comparisons required to determine k_{Cl_2}* are at the extreme ends of our calibrated spectral response curves. Furthermore, it is likely that some KrCl* emission extends into the v.u.v. and, therefore, is not included in I_{KrCl}*. Careful recalibration experiments now under way will decrease the uncertainty of this branching ratio. The lack of reliable oscillator strengths for transitions in

in the 6s-6p manifold precludes a determination of k_{XeCl} by the method outlined above. However, evidence will be presented that supports the assignment of unity branching ratio for excimer formation from Xe* + Cl_2.

Extensive vibrational relaxation of the excimers occur in the 1-40 Torr pressure range as shown in Figure 1 for XeF from Xe* + ONF. Similar results were obtained for XeCl*, KrCl* and KrF*. Since significant relaxation is evident even at < 5 Torr, simple consideration of the gas-kinetic collision frequency at these pressures ($\sim 4.3 \times 10^7$ at 5 Torr) suggest a lifetime > 50 ns for the rare gas-halide excimers. Figure 1 also shows the XeF emission at $\sim$ 265 nm, originating from an upper excimer excited state ($^2\pi_{1/2}$) that correlated with the ($^2P_{1/2}$) state of the rare gas ion. The state responsible for the main excimer emission correlates with the $^2P_{3/2}$ ion state. Emission from this upper state was also identified for XeCl*, KrF* and KrCl*. The separation between the maximum of the primary emission and that due to this new state is 9422 cm^{-1} (1.17 eV) for XeF, 9867 cm^{-1} (1.22 eV) for XeCl, 5050 cm^{-1} (0.63 eV) for KrF and 5380 cm^{-1} (0.67 eV) for KrCl. The dynamics of these reactions as deduced from the emission spectra of the excimers will be briefly discussed in terms of analogy to alkali metal atom reactions.

References:

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TABLE II. Rate Constant Summary

Reagent	k_Q (10^{11} cm^3 molec^{-1} sec^{-1})			Relative k_{KrX}	Relative k_{XeX}
	$Ar(^3P_2)$	$Kr(^3P_2)$	$Xe(^3P_2)$	k_{KrCl}	k_{XeCl}
Cl_2	771	773	772	100[a]	100[a]
F_2	775	772	775	100	100
Br_2	65		60		
ClF_3				62[b]	33[b]
ICl	61		50		
OF_2	57	53	57	82	80
CF_3OF	43	42	47	35	25
SF_6	16		28	≈0	≈0
SeF_6	71		65	0	0
TeF_6	58		63	0	0
SO_2F_2	42			0	0
NF_3	14	16	9	7	14
N_2F_4	31	33		22	27
NOF	36	47	30	<5 [c]	40; (43)[d]
NOCl	48		51		
HCl	37		56		
HBr	57		61		
HI	75				
CF_3I			52		
BrF_5				0; (33)[e]	0; (23)[e]

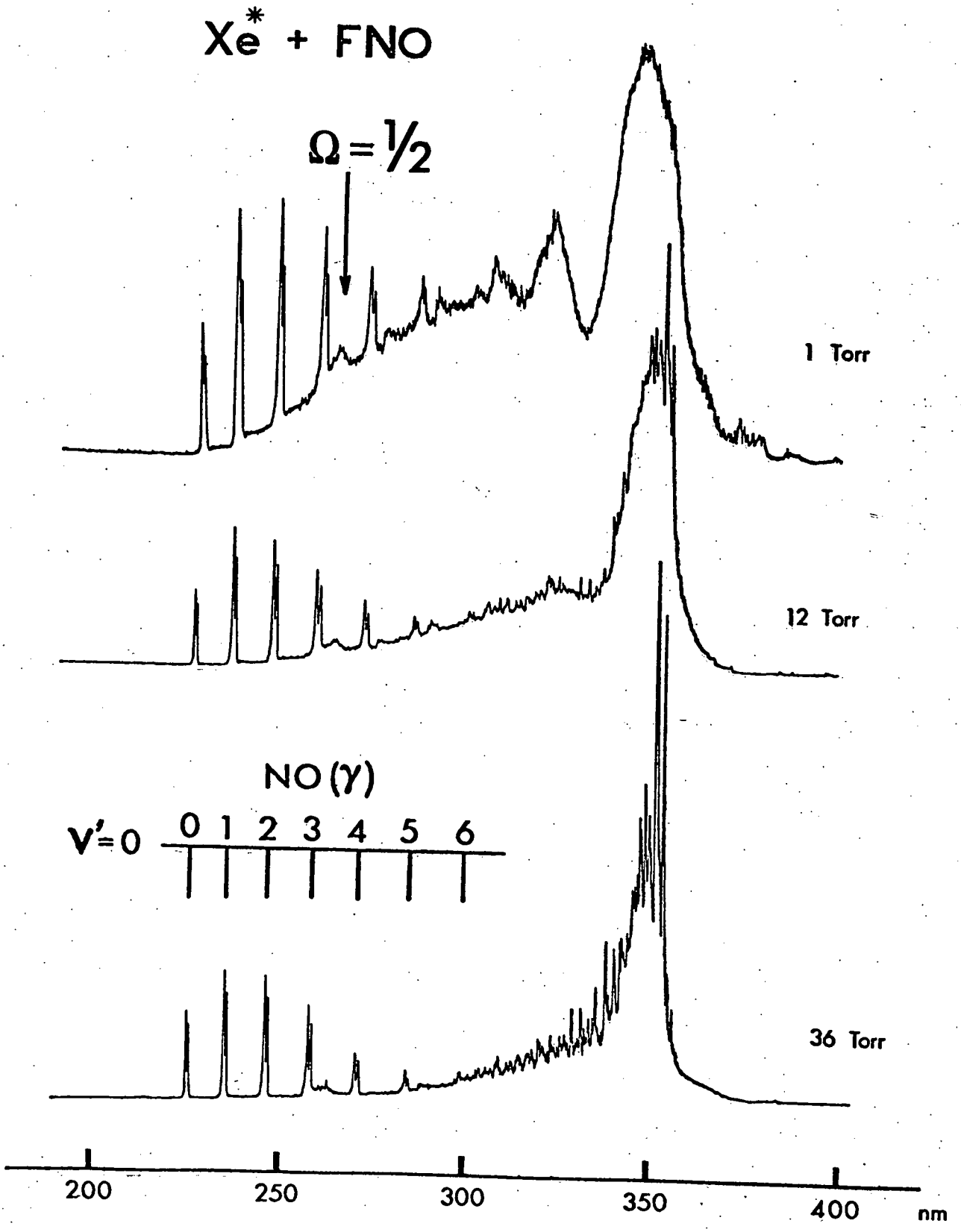
a. Cl_2 was used as the reference reaction for assigning k_{MX}.

b. Both chloride and fluoride excimer emission was observed.

c. No (γ) and (β) emissions were the main product. This value is based on an estimate of the KrF emission intensity.

d. No (γ) bands were also observed. This value includes the contribution of the γ-bands.

e. MBr rather than MF was the product.


Xe* + FNO
Ω = ½
1 Torr
12 Torr
NO(γ)
V'=0
0 1 2 3 4 5 6
36 Torr
200
250
300
350
400
nm